Towards a Constrained Understanding of Europa's Subsurface Driven By Magnetic Induction Sounding

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1 Introduction

Jupiter's moon Europa (radius $R_E = 1,560.8$ km) orbits on a nearly-circular orbit at approximately $9.38 R_J$ from the giant planet ($R_J = 71,492$). The moon's orbit is contained to the Jovigraphic equatorial plane, and the planetary rotation overtakes the moon's motion once about every 10 hours. The magnetospheric fields and plasma contained to the Jovian magnetosphere, most dense near the magnetic centrifugal equator (Phipps & Bagenal, 2021), are convected with the rotation of the planet carried by currents coupling to Jupiter's ionosphere (Hill, 1979). This causes the moon to be continuously overtaken by a corotating Jovian magnetic field and thermal plasma population (Bagenal & Delamere, 2011; Bagenal et al., 2015). The moon also possesses a dilute, molecular oxygen-dominated atmosphere (also containing water group products; see, e.g., Saur et al., 1998; Volwerk et al., 2001; Roth et al., 2016; Plainaki et al., 2018; Roth, 2021) with column densities of roughly 10^{18} cm^{-2} (Hall et al., 1995, 1998). This atmosphere interacts with the impinging plasma to produce a complex system of currents that drive perturbations in the electromagnetic fields in the vicinity of the moon (i.e., draping; see, e.g., Kivelson et al., 1996, 2009; Rubin et al., 2015; Simon et al., 2021; Harris et al., 2021; Haynes et al., 2023; Addison et al., 2024). This also produces a system of nonlinear standing Alfvén currents that stretch hundreds of R_J and connect to Jupiter's polar ionosphere (Neubauer, 1980, 1998), generating measurable auroral signatures (Bonfond et al., 2017; Bhattacharyya et al., 2018; Hue et al., 2023).

Jupiter's magnetic axis is inclined by 9.6° against its polar axis. Hence, at the (Jovian) synodic rotation period $T_s = 11.23$ hr, the icy moon is subject to an undulation in the corotating magnetospheric fields and plasma density. This well-behaved, continuous oscillation can be described in Europa's reference frame as an ambient field with a constant southward component and a time-varying "horizontal" component (in the equatorial plane). The well-behaved time variability of the Jovian background field sweeping over Europa at Jupiter's synodic period drives a corresponding, time-varying induced field in the conductive layers of the moon. Other driving frequencies include the orbital period ($T_o = 85.2$ hr), beats of T_s, T_o, and harmonics of these periods. However, these secondary driving signals are at least an order of magnitude weaker than T_s, if not more. This induction signal was discovered above Europa's surface in magnetometer data taken by the Galileo spacecraft (Kivelson et al., 1999, 2000), which visited the moon and the Jovian system in the late '90s and early 2000's. In principle, this could stem from conduction in a salty, subsurface ocean *or* a very dense ionosphere (Neubauer, 1998; Hartkorn & Saur, 2017). However, several studies have found with confidence that this signal stems from induction driven by a liquid water subsurface ocean beneath Europa's icy crust (Zimmer et al., 2000; Spohn & Schubert, 2003; Kivelson et al., 2004, 2009; Schilling et al., 2007; Seufert et al., 2011; Addison et al., 2024). The time-variability in Europa's upstream magnetospheric field vector and thermal plasma density cause the perturbations due to plasma currents to also vary at the period T_s. The oscillation in the plasma currents produces only a marginal modification to the inductive response (Schilling et al., 2007; Neubauer,

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1998), and is often neglected in studies of magnetic induction in Europa's ocean (Zimmer et al., 2000; Seufert et al., 2011; Styczinski & Harnett, 2021a; Vance et al., 2021a).

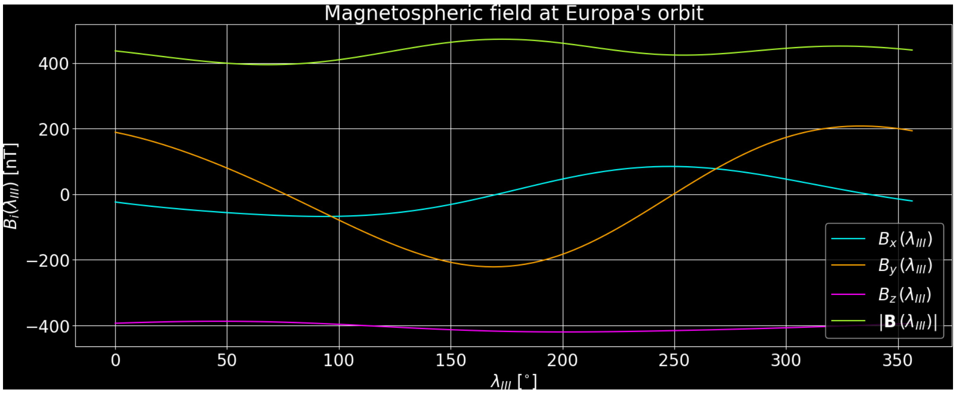

Figure 1. Time-varying components of the ambient Jovian magnetospheric field $\mathbf{B}_0$ at Europa, given as a function of (West) longitude around the planet. Magnetic field magnitude $|\mathbf{B}_0|$ is shown in yellow, and the $x-$, $y-$, and $z-$ components of the ambient field in the Satellite Interaction System (see section 2) are shown in teal, orange, and magenta, respectively. System III Longitude λ_{III} increases moving clockwise around Jupiter.

Europa is internally differentiated (Anderson et al., 1997; Schubert et al., 2004), with its subsurface distributed (moving radially outward) into a metallic core, a silicate rock mantle, and a salty liquid water ocean $\sim 50-200$ km thick, all under an icy crust (e.g., Anderson et al., 1998; Kivelson et al., 2000; Zimmer et al., 2000). The surface is estimated to be $10-90$ Myr old from it's cratering morphology (Kargel et al., 2000), relatively young for most solar system bodies. Europa is one of the smoothest bodies in the solar system. Its subtle morphological variation across the surface takes the form of dark-colored lineaments and stratigraphically young lineae, forming overlapping patterns of grooves and ridges (e.g., Helfenstein et al., 1998). The surface is primarily water ice, both in the amorphous and crystalline states (Hansen & McCord, 2004; Paranicas et al., 2018). A rich coupling between the moon's exterior magnetospheric environment and interior geochemical transport results in the presence of hydrated sulfates and chlorinated salts across Europa's surface, as well as presumably occupying the moon's ocean and supplying part of its conductivity (Johnson, 1990; Carlson et al., 1999; Cassidy et al., 2013; Ligier et al., 2016; Vance et al., 2018; Trumbo & Brown, 2023; Addison et al., 2023). Brines in the ocean are highly conductive compared to the insulating ice in the crust and mantle (Vance et al., 2021b). Since its discovery, Europa's ocean has been a desireable candidate to house extraterrestrial life (Vance et al., 2018; Howell & Pappalardo, 2020). This is primarily driven by its ability to maintain a tremendous volume of water in the liquid state beneath its icy crust. Europa can do so due to its orbital eccentricity, maintained by mean-motion resonance with the other inner Galilean moons Io and Ganymede (Walker & Rhoden, 2022). Furthermore, the evidence is mounting for extensive geochemical interactions between Europa's warm mantle/core, its ocean, and its icy surface (Cassidy et al., 2013; Trumbo et al., 2020; Trumbo & Brown, 2023; Mermy et al., 2023). In this context, brines in the ocean are highly conductive compared to the insulating ice in the crust and mantle (Vance et al., 2021b).

To date, the approaches used to constrain Europa's subsurface distribution through the inductive signals from the ocean have considered only forward-modeling analysis or implemented, at most, a least-squares approach (e.g., Kivelson et al., 2002). An analogous inversion for the nature of Ganymede's internal and induced fields, using data obtained by the Juno, the most recent spacecraft to visit the Jovian system, has only slightly increased the complexity by using a weighted least-squares approach (Weber et al., 2022). However, this more updated approach has not been carried out for the Juno flyby of Europa. Hence, there is substantial room for improvement in the breadth and depth of these approaches to further elucidate the stratification of Europa's upper subsurface (Vance et al., 2018). With the discovery of Europa's putative liquid water ocean, the question of how Europa's outer subsurface is composed and distributed arises naturally. The presence of a large volume of liquid water combined with the rich geochemical system places Europa as a promising solar system candidate for biological habitability (Vance et al., 2018; Howell & Pappalardo, 2020). Constraining the subsurface of Europa is an overarching goal of the en-route Europa Clipper investigation, a Flagship-class NASA Mission set to encounter the moon across many grazing flybys, with closest approaches reaching down to $\mathcal{O}(10\text{ km})$ (Cangahuala et al., 2025). In this work, we seek to constrain the depth, thickness, and composition of Europa's interior ocean, all of which determine the radial conductivity profile. By revealing the likely location and conductivity (i.e., composition) of Europa's subsurface ocean, we can provide refined understanding that will supplement the Europa Clipper mission objectives and further our understanding of the enigmatic icy moon.

2 Model Description

Throughout this study, we employ the Cartesian Satellite Interaction System (SIS) with coordinates $\mathbf{r} = (x, y, z)$ at each moon. This system originates at the center of Europa with the following orthonormal basis vectors: $\hat{\mathbf{x}}$ is aligned with the direction of corotational magnetospheric flow, $\hat{\mathbf{y}}$ is aligned with Jupiter's spin axis, and $\hat{\mathbf{z}}$ bears roughly toward Jupiter, completing the right-handed system.

2.1 Model of Europa's Inductive Response

As discussed more extensively in the midterm report, as well as in, e.g., Zimmer et al. (2000) and Vance et al. (2021b), it is possible to obtain an analytical representation of Europa's inductive response in three dimensions with some simplifying assumptions. While not exhaustive, we provide several of the simplifying assumptions that yield the below formalism, which we will also carry in our work. These assumptions are as follows:

1. That the Jovian magnetospheric field is homogeneous on the scale of Europa's body. This is true to within a high degree (several orders of magnitude) of precision, as the Jovian field changes by ≈ 3 nT, or less than 0.8% (Connerney et al., 2020, 2022).
2. That there are no hydrodynamic convective currents in Europa's subsurface ocean that may trigger motional induction. This contribution would only be a minor modification $\approx 1 - 2$ nT (see also Šachl et al., 2025).
3. That the ocean is not located deep (i.e., a small fraction of R_E) within the insulating icy "crust" (Kivelson et al., 1999; Vance et al., 2018).
4. That the degree of anisotropy is weak, that is, contributions from a non-spherical annular shape are minor. Estimates place this contribution at no more than 2 nT near Europa's surface (Styczinski & Harnett, 2021a).
5. That the influence of thermal gradients is weak enough to not drive currents and to not substantially impact the conductivity itself (Vance et al., 2021a).

Assumptions (4) and (5) are not requisite to obtain an analytical solution for the inductive response, but we neglect these effects in analogy with other studies of induction sounding (e.g., Strack & Saur, 2024).

The inductive response of Europa with a specified subsurface conductivity (see Figure 2) is obtained by solving the diffusion equation:

$$\nabla^2 \mathbf{B} = \mu_0 \sigma \, \partial_t \mathbf{B} \quad ; \quad \text{if } \sigma \longrightarrow 0 \quad \Longrightarrow \quad \nabla^2 \mathbf{B} = 0 \quad . \tag{1}$$

By applying the above assumptions and appropriate boundary conditions, the inductive response can be described as a dipole magnetic field with a moment oriented opposite to the inducing field (Zimmer et al., 2000; Rubin et al., 2015; Styczinski & Harnett, 2021b). This set of assumptions yields the (time-varying) induced magnetic dipole moment $\mathbf{M}_{\text{ind}}$ possessed by Europa, centered at its core and given by (for external oscillation frequency ω)

$$\mathbf{M}_{\text{ind}} = -A\, e^{-i(\omega t-\phi)}\, \frac{2\pi R_E^3}{\mu_0}\, \mathbf{B}_{h,0} \qquad . \tag{2}$$

Here, $\mathbf{B}_h$ is the horizontal component (i.e., the component projected onto the $\hat{\mathbf{x}}$–$\hat{\mathbf{y}}$ plane) of the external Jovian field $\mathbf{B}_0$ (see Figure 1). For a perfect conductor, the value of A approaches unity and the phase lag ϕ vanishes. The negative sign in equation (2) represents the principle of Lenz's law, that the induced field directly opposes the primary magnetospheric inducing field. A finite conductivity of the ocean layer results in modified values for A and ϕ. The field at a given position $\mathbf{r}$ is therefore given by the expression

$$\mathbf{B}_{\text{ind}} = \frac{\mu_0}{4\pi}\left(3(\mathbf{r}\cdot\mathbf{M}_{\text{ind}})\mathbf{r} - \mathbf{r}^2\mathbf{M}_{\text{ind}}\right)/r^5 \qquad . \tag{3}$$

Equation (3) will be used to relate the spacecraft coordinates and the expected magnetic field measurements for a given conductivity profile in Europa's subsurface.

In order to ensure our inversion is tractable, we assume that Europa's subsurface ocean possesses a constant conductivity. This may produce errors on the order of $1-10$ nT in the induced field at a given point at or near Europa's surface, but this effect falls off with the same radial behavior as the induced field itself (i.e., $1/r^3$). As discussed in the midterm report, the parameters A and ϕ are expressed in terms of σ_o, r_1, r_0. Following Zimmer et al. (2000) and Parkinson (1983), using Bessel functions of the first kind J_m (of order m), these relations read

$$Ae^{i\phi} = \left(\frac{r_0}{R_E}\right)^3 \frac{CJ_{\frac{5}{2}}(r_0\,k) - J_{-\frac{5}{2}}(r_0\,k)}{CJ_{\frac{1}{2}}(r_0\,k) - J_{-\frac{1}{2}}(r_0\,k)} \qquad , \tag{4}$$

where

$$C \equiv \frac{r_1 k J_{-\frac{5}{2}}(r_1\,k)}{3J_{\frac{3}{2}}(r_1\,k) - r_1 k J_{\frac{1}{2}}(r_1\,k)} \qquad . \tag{5}$$

In equations (4) and (5), the complex wavenumber parameter

$$k = (1-i)\sqrt{\frac{\mu_0\sigma_o\,\omega}{2}} = (1-i)\sqrt{\frac{\pi\mu_0\sigma_o}{T_s}} \tag{6}$$

is dependent on the conductivity and driving frequency. We assume that the driving frequency is the synodic rotation frequency $1/T_s$ *in isolation*, that is, we neglect $1/T_o$ as well as any harmonics and beats of these frequencies. This is motivated by their relative weakness, 1–3 orders of magnitude below the synodic signal (see also Vance et al., 2021b). We assume that the magnetic permeability μ in Europa's subsurface is equal to its value in a vacuum μ_0 (Vance et al., 2021b).

2.2 Inversion Model to Constrain Europa's Interior

2.2.1 Model Discretization

Following the approach described in our midterm report and applied in many previous studies of induction at Europa (e.g., Zimmer et al., 2000; Kivelson et al., 1999), we assume that the moon's subsurface stratification is described with the following layers. Each layer of the moon is spherically symmetric. The subsurface is distributed into a core/mantle with conductivity $\sigma = 0$, an ocean layer of conductivity σ_o with inner radius r_1 and outer radius r_0, and an insulating crust with $\sigma = 0$. This structure is illustrated in

Figure 2. While some studies have taken the approach of including a finite conductivity of the mantle region too (e.g, Schilling et al., 2007), this conductivity value is often much weaker than that of the ocean. Furthermore, the skin depth

$$\delta = \sqrt{\frac{2T_s}{\sigma\mu}} \tag{7}$$

is only approximately δ=250 km for a conductivity of 1 S/m.

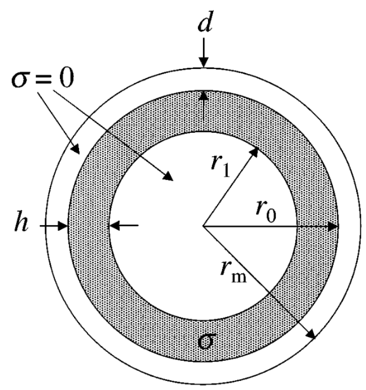

Figure 2. Radially symmetric structure assumed for Europa's subsurface. Adapted from (Zimmer et al., 2000).

For a value of $\sigma = 10$ S/m, it becomes only 80 km. That is, the skin depth for T_s around the range of expected conductivities is the same order of magnitude as the thickness / depth of Europa's ocean. Any conducting layer that is thicker than the skin depth will "screen" the interior from any induction signal. Even if the skin depth is the same order as the ocean thickness, the signal will be attenuated in strength before it reaches Europa's mantle. In fact, this property can even partially obscure the lower boundary of Europa's ocean (Schilling et al., 2004; Schilling et al., 2007). Hence, we retain the assumption of $\sigma = 0$ in Europa's mantle.

In addition, we assume that the phase lag in Europa's ocean is negligible. The phase lag corresponds to twice the propagation time for the electromagnetic signal to reach the inducing regions in Europa's subsurface: that is, the time it takes for a given background field $\mathbf{B}_0$ to result in an inductive response for that given field outside of Europa's solid body. The phase lag often is assumed or obtained to be on the order of 5° or less at Europa (e.g., Zimmer et al., 2000; Strack & Saur, 2024; Cervantes et al., 2025). This corresponds to only a marginal change in the magnetospheric background field at Europa. As will be shown by the results (see section 3), the assumed model for Europa's subsurface is highly correlated between parameters without including this variable. For this reason, we fix the phase lag to 0° for this study and leave the decoupling of the phase lag from the other parameters driving induced field signatures a subject of future work.

This discretization leaves us with three free parameters to describe the ocean our model, for a fixed value of the phase lag (e.g., Seufert et al., 2011). These are the inner thickness r_1, the outer thickness r_0, and the conductivity σ. The difference $r_0 - r_1$ defines the ocean thickness, and $R_E - r_0$ defines the ocean depth (see Figure 2). This is equivalent to the discretization used to obtain the forward model in our approach. Therefore, our model for Europa's subsurface includes the following three parameters, in vector form:

$$\mathbf{m} = [r_1, r_0, \sigma] \quad . \tag{8}$$

The quantities contained inside $\mathbf{m}$ will be the target of each inversion strategy we apply to constrain Europa's interior through magnetic induction sounding. We describe such methods below.

In the following, we compactly review the mathematical formalism that describes each inversion strategy. For a more detailed description, we refer the reader to the lecture notes and to the Aster et al. (2013) textbook on Inverse Methods (Parameter Estimation and Inverse Problems). We also describe any specific choices made for the implementation of our software. Each model is written in python for clarity and access to advanced computational (e.g., `scipy`), array-manipulation (e.g., `numpy`), and plotting (e.g., `matplotlib.pyplot`) libraries that are pre-built and readily imported. While we do apply these libraries, we have produced all code that performs the routines from scratch, excluding only the highly optimized implementation of the Levenberg-Marquardt Trust Region Reflective algorithm (described in section 2.2.3).

The algorithms we have developed work by calling functions that each evaluate specific steps of the routine. The first helper function calculates the actual dipole moment $\mathbf{M}$ from a given input model parameter set $\mathbf{m}$ using equations (3)-(6). Then, another one takes in this dipole moment along with a set of positions and a given *inducing* field $\mathbf{B}_h$ in order to calculate the resultant magnetic field vectors along the trajectory. These functions combined is equivalent to evaluating $G(\mathbf{m})$. We then employ another helper function to evaluate the objective function (i.e., equation 9) by stacking the observed magnetic field vectors into a single dimension in the same way as the observations, subtracting them element-wise, and calculating the residual norm. Further details on the specifics of our implementation are omitted here, but we provide insights when necessary for interpretation of the results.

2.2.2 Inversion Algorithm: Gauss-Newton

We begin by describing the Gauss-Newton method for determining a solution $\mathbf{m}$ to the objective function given by the sum of weighted, squared residuals. That is, we seek to find the model $\mathbf{m}$ that minimizes the scalar valued function

$$f(\mathbf{m}) = \sum_{i=1}^{M} \left(\frac{G(\mathbf{m})_i - d_i}{s_i} \right)^2 \qquad . \tag{9}$$

In this objective function, we have retained the "standard" nomenclature from Aster et al. (2013), where G represents the vector-valued operator that converts the 3 dimensional model vector to the predictions for data at each location of a measurement:

$$G(\mathbf{m}) = \mathbf{d} \qquad .$$

The values of the data are given by the vector $\mathbf{d}$, with M distinct elements. For data that are given by magnetic field vectors themselves, we perform a stacking operation to rasterize the data into a single vector. The standard deviation s_i for each datapoint d_i is distinct from the common nomenclature for standard deviation to avoid confusion with the conductivity parameter.

The method is akin to Newton's method, but generalized to allow for a non-square G operator. The method attempts to find the set of parameters that minimize the weighted residual norm in equation (9). By defining a vector $\mathbf{F}(\mathbf{m})$ that contains M elements of the weighted residuals for each datapoint, we can write

$$\nabla f(\mathbf{m}) = 2\mathbb{J}(\mathbf{m}) \cdot \mathbf{F}(\mathbf{m}) \qquad ,$$

where $\mathbb{J}$ is the Jacobian given by

$$\mathbb{J}(\mathbf{m}) = \frac{1}{2} \left(\nabla f(\mathbf{m}) \cdot \mathbf{F}^{-1}(\mathbf{m}) \right)^{\mathrm{T}} \qquad .$$

We can similarly define the Hessian $\mathbb{H}$ for a given model $\mathbf{m}$ as

$$\mathbb{H}(\mathbf{m}) \approx 2\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbb{J}(\mathbf{m}) \qquad .$$

This approximation neglects the term $\mathbb{Q}$ that is considered small in the realm of validity of this model (Aster et al., 2013). Hence, combining these definitions with some manipulation of the above yields an update for $\mathbf{m}$, denoted $\Delta\mathbf{m}$. This method essentially seeks the minimum of equation (9) by differentiating and equating to zero, and stepping along the direction towards the nearest minimum in model parameter space. Following equations 9.13-9.28 in Aster et al. (2013), the update is given by the solution to

$$\left(\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbb{J}(\mathbf{m}) \right) \cdot \Delta\mathbf{m} = -\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbf{F}(\mathbf{m}) \qquad .$$

The updated model after an iteration is then equal to

$$\mathbf{m}_{\text{new}} = \mathbf{m} + \Delta\mathbf{m} \qquad .$$

To implement this approach, we calculate the Jacobian using finite differencing. This method incorporates a step through parameter space that, due to its dependence on the data residuals and forward model, does not permit bounding. Additionally, the method is susceptible to finding local minima or convex extrema (saddle points) in parameter space.

2.2.3 Inversion Algorithm: Levenberg-Marquardt (Trust Region Reflective)

To improve upon our approach detailed in section 2.2.2, we also implement the Levenberg-Marquardt algorithm for inversion. This follows the same mathematical rationale as Gauss-Newton. However, it includes a balance between the Gauss-Newton step and a "blind", steepest descent step. This is achieved through the update including a term identical to the Gauss-Newton update, and a term $\lambda\mathbb{I}$ proportional to the identity matrix that is equivalent to a step using steepest descent. The update is given by

$$\left(\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbb{J}(\mathbf{m}) + \lambda\mathbb{I}\right) \cdot \Delta\mathbf{m} = -\mathbb{J}(\mathbf{m})^{\mathrm{T}} \cdot \mathbf{F}(\mathbf{m}) \qquad . \tag{10}$$

The tunable weighting parameter λ controls the balance between the two methods.

To implement this approach, we rely on a highly robust pre-defined function in the `scipy.optimize` library named `least-squares`. This inversion method uses a modified version of the Levenberg-Marquardt algorithm called "Trust Region Reflective". The Levenberg-Marquardt approach performs a similar update as described in equation (10), following the same Newton-step method approach as Levenberg-Marquardt. However, the Trust Region Reflective modification to the algorithm is able to "safely" navigate any regions of negative curvature in the objective function minimization. This means that if the input "guess" for $\mathbf{m}$ is far from the minimizing solution, i.e., if $\mathbf{m}$ needs a large update, the Trust Region Reflective approach will quickly push it out of this location in parameter space, whereas the Levenberg-Marquardt approach may require many iterations, get stuck, or approach nearby local saddle points. This algorithm also takes in bounds as an argument and imposes these domains on the parameters as they are evolved, which facilitates incorporating known constraints to guide the method only through the "allowed" regions of parameter space. Both the original Levenberg-Marquardt and Trust Region Reflective do not use a constant, set value of λ but rather update it based on the residuals and behavior of the objective function / cost function.

2.2.4 Inversion Algorithm: Metropolis-Hastings Markov Chain Monte Carlo

The final method we employ in this study is a Markov Chain Monte Carlo (MCMC) approach called Metropolis Hastings. This method is different from the other two in that it is a statistical approach: rather than providing a path to the single "best" model, such a strategy yields information about the probability distribution of the model parameters in parameter space. This is in the form of a posterior distribution that is evolved through iterative sampling. Such methods provide a means to garner more information about parameter space than a single inversion minimization or optimization routine, since the statistical information can be used to glean information about the physical model parameters.

The Metropolis Hastings algorithm calculates a sequence of random variables where each sample m_i is drawn from a transitional probability

$$\mathbb{P}(m_i|m_{i-1})$$

which depends solely on the previous sample m_{i-1}. The samples are normally distributed around the variances s^2:

$$m_i = m_{i-1} + \eta_{i-1} \qquad ; \qquad \eta_i \sim \mathcal{N}(0, s^2) \quad , \tag{11}$$

where $\mathcal{N}(0, s^2)$ is used to denote a normal distribution with mean of zero and a standard deviation of s. Then, $\mathbb{P}(m_i|m_{i-1}) = \mathcal{N}(m_{i-1}, s^2)$, where $\mathbb{P}$ is the proposal distribution. The MCMC algorithms rely on the fact that the distribution of samples given this proposal distribution will limit to the true posterior distribution of all model parameters.

The Metropolis-Hastings MCMC routine takes the parameter guess that starts the Markov Chain as an input. It also takes in the step sizes (s^2 for each model parameter for the normal distribution sampling), as well as the prior distribution for each model parameter prescribed by the operator. We opt to use exclusively a uniform prior distribution, since there is not confidence enough in Europa's subsurface stratification to truly center a normal distribution around one particular value for each element of $\mathbf{m}$, that is, the inner and outer radii or the conductivity. Hence, our prior serves essentially as a domain limiter in parameter space, where the likelihood will be zero outside of the range and a constant value inside. The algorithm is constructed and implemented as follows:

1. Propose a new sample (i.e., a model) m_i' by drawing from a proposal distribution $\mathbb{P}(\,\cdot\,|m_{i-1})$. This is just the conditional probability of an unknown variable (i.e., the one to be selected) conditioned on the previous sample m_{i-1}.
2. Calculate the acceptance probability term
$$A = \frac{\pi(m_i')q(m_{i-1}|m_i')}{\pi(m_{i-1}')q(m_i|m_i')} \qquad .$$
In this expression, the nomenclature is adopted from that of the class lecture. The prime represents the fact that the new sample m_i' may *not* be accepted, even if it has a high likelihood, and that means it would not contribute to the calculated posterior distribution.
3. Accept sample with probability α, that is,
$$\alpha = \min(1, A) \qquad ,$$
where there is a probability α that the proposed sample is accepted and $1 - \alpha$ that the sample is not stored.
4. Repeat steps (1-3).

This process obtains an empirical distribution of saved states that will approach the posterior probability distribution of the model parameters given the available data.

While the statistical methods boast the ability to constrain far more information about the model parameters and their likelihoods than a deterministic method, the Metropolis-Hastings algorithm has a few important caveats. First of all, the method results in samples which are inherently autocorrelated (i.e., slightly biased compared to the true distribution). That occurs because each sample is selected using probabilities evaluated using the previous model only. Hence, locally on the Markov Chain the samples are not normally distributed. Also, the fact that an initial sample prescribes the starting state of the Markov Chain implies that some amount of samples near the onset of the algorithm (e.g., the first 10%–20% of iterations) must be recycled (i.e., ignored) to ensure that there is minimal bias introduced from the choice of starting guess. This is referred to as the *burn-in* period.

In order to balance the benefit of a large sample number against the limited computational power of a single desktop machine, we perform runs with $400,000$ and $800,000$ samples. We discard the first $\approx 100,000$ as burn-in. As we will demonstrate in section

3.2.1, the sensitivity of the inversion to the conductivity is substantially higher than that of either radius (but has a similar sensitivity to the thickness, phenomenologically). For this reason, we treat the third parameter in the model vector as $\log \sigma$ instead, so that its variability occurs on smaller steps. This form of regularization allows us to circumvent the disparity in sensitivity of our model between either the inner or outer radius and the conductivity value of Europa's subsurface ocean. Our step sizes for the three parameters in $\mathbf{m}$, translated to the variances of the distribution from which they are sampled (see equation 11), are given by $[0.3, 0.16, 0.02]$, for the inner radius (in km), outer radius (in km), and the logarithm of the conductivity (in log S/m).

3 Results and Discussion

3.1 Results For Synthetic Data

In order to validate and test our chosen strategies of recovering the depth, thickness, and conductivity of Europa's subsurface ocean from Galileo magnetometer data (see section 2), we apply our suite of inversion algorithms to a synthetic dataset. To do so, we apply an assumed background magnetospheric field given Europa's position with respect to Jupiter's plasma sheet using the Juno magnetospheric field model (Connerney et al., 2020, 2022). Since position with respect to the Jovian plasma sheet corresponds to a longitude in the system III coordinates of the Jovian magnetosphere (e.g., Bagenal et al., 2016; Phipps & Bagenal, 2021; Addison et al., 2023), we can readily determine the background magnetospheric field vector from an assumed position of the moon. This synthetic data is generated under the following assumptions.

First, we let Europa lie at maximum distance below the Jovian magnetospheric plasma sheet. This corresponds to a background field vector of $\mathbf{B}_0 = [4, 209, -385]$ nT using the Juno model (Addison et al., 2023; Haynes et al., 2023). It is at this location, as well as the correspondent position of the moon at maximum distance above the center of Jupiter's plasma sheet, that the induction signal expected at the moon is the strongest. This is because the magnitude of the *inducing* component of the Jovian background field, i.e., the non-southward, or horizontal ($\mathbf{B}_h$) component, is maximized for this configuration. Analogously, when the moon is at the center of the Jovian plasma sheet, the magnetospheric field vector is its *closest* to being oriented perfectly southward. The time-varying component of the magnetospheric field (at the period T_s) is contained in the plane perpendicular to Jupiter's spin axis, which is aligned with the $(+z)$ coordinate of our Satellite Interaction System.

Then, using our assumed configuration of the moon, we produce a synthetic flyby trajectory along which to sample the "expected" induced field at Europa, given an assumed model of the interior $\mathbf{m}$. Our trajectory travels along the $(+x)$-direction of our coordinate system, for simplicity of interpreting our synthetic flyby results. Our assumed trajectory, for a parameter s can be described as a line in space given by

$$\mathbf{x}(s) = [-5,000 + s, 100, R_E + 100] \text{ km} \qquad . \tag{12}$$

Hence, the spacecraft that would observe such data would perform a flyby with closest approach (CA) above Europa's north pole, traveling from upstream to downstream (i.e., along the direction of Europa's orbital motion). In order to provide a nonzero reading for the background field in the z-component, we incorporate a small offset between the trajectory and the x-axis in the $z = 0$ plane. That is, we add a small (100 km) value to the y position at all points.

We make our results quantitatively realistic by adding noise in the form of random fluctuations. The noise level is selected to be ~ 2 nT, which is comparable to the values of the fluctuations visible in the actual flyby data (see also Figure 4). Finally, we fix an assumed model of Europa's interior. We select the values of $r_1 = 1,400$ km and

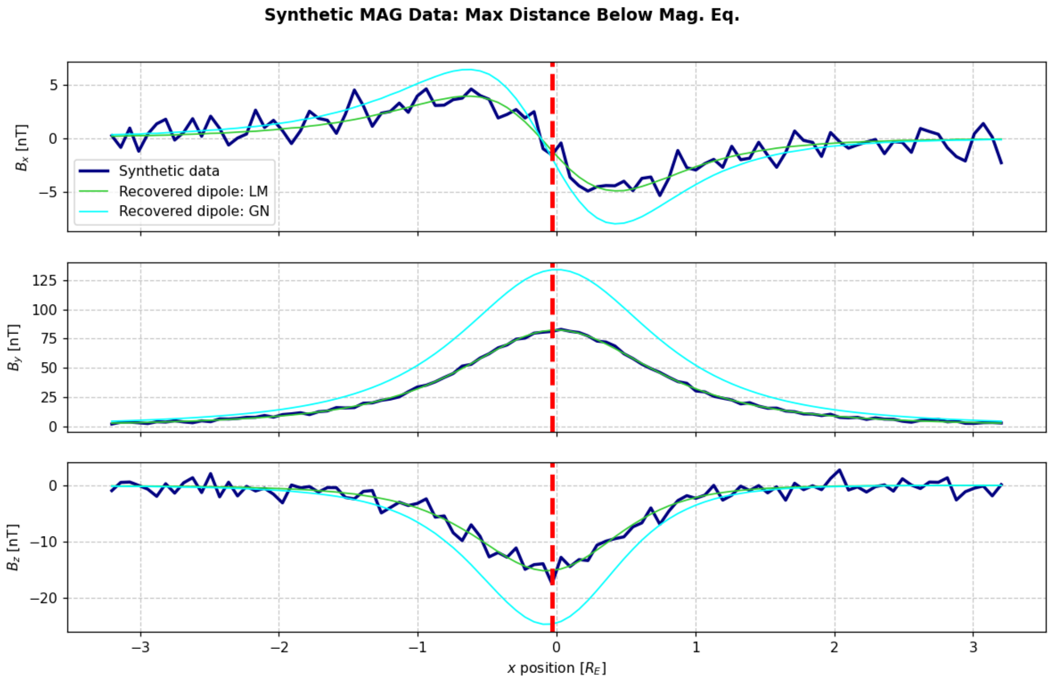

Figure 3. Synthetic data and deterministic inversion model results. The synthetic flyby data, shown in navy, is detrended (i.e., shown with $\mathbf{B}_0$ subtracted), so the only signal visible is the induced field used to create it (model vector of $\mathbf{m}_{\text{true}} = [1400, 1540, 2.5]$) and the 2 nT Gaussian noise added to the data to simulate nonstationarities in the ambient electromagnetic fields near Europa. The induced dipole resulting from the parameters recovered with Levenberg-Marquardt is shown in lime green, and the induced dipole from the parameters obtained with Gauss-Newton is shown in cyan. The horizontal axis is labeled with x-position of the hypothetical spacecraft, and the magnetic field components are displayed in nT (vertical axis).

$r_2 = 1,540$ km for the inner and outer radii of the ocean, respectively. This yields an assumed thickness of 140 km, a within the range of uncertainty in constraints for the moon's interior obtained through previous analyses (e.g., Zimmer et al., 2000). The conductivity assumed is $\sigma = 2$ S/m, yielding an assumed model vector of $\mathbf{m}_{\text{true}} = [1400, 1540, 2.5]$. Since the Jovian background field for this configuration has only a nontrivial component in the y-direction, this chosen geometry for the trajectory facilitates the interpretation of our results by providing straightforward access to the involved physics. Furthermore, by assuming that the inducing field is the strongest, we essentially give our models the "best case" scenario for being able to recover the data.

Our results obtained from applying the deterministic inversion models (see sections 2.2.2 and 2.2.3) are depicted in Figure 3. The data is captured along a line parallel to the x-axis as described above, and thus is labeled according to the x displacement rather than time for simplicity. As can be seen by the comparison between model output and synthetic data, neither fit is excellent. However, there are still many elements of the signal that are recovered by the Gauss-Newton model. However, the Levenberg-Marquardt algorithm with the Trust Region Reflective modification performs extremely well. It is immediately clear that the orientation and shape of the induced dipole moment provided by the two algorithms are correct: the sign and behavior of each component follows that of the synthetic data nearly precisely. The recovered model parameters for the two fits are as follows:

- $\mathbf{m}_{\text{G-N}} = [r_1 = 1309.04 \text{ km}, r_0 = 1515.82 \text{ km}, \sigma = 2.279 \text{ S/m}]$
- $\mathbf{m}_{\text{L-M}} = [r_1 = 1300.46 \text{ km}, r_0 = 1350.41 \text{ km}, \sigma = 2.69 \text{ S/m}]$

The Gauss-Newton fit does not perform nearly as well as the Levenberg-Marquardt routine, but it still is able to quantitatively replicate a large fraction of the total signal. The B_x component is nearly perfectly matched, and this is the weakest one. In addition, B_z component is also nearly reconciled, but it is slightly overestimated around closest approach where the induced dipole signatures are the strongest. The B_y component has the most difficulty with the model parameters (cyan curve in Figure 3). For the Gauss-Newton fit, the residual norm is approximately $||res|| \approx 180 - 200$. This is a substantial improvement from the initial guess of $\mathbf{m}_0 = [1300.0, 1560.0, 1.]$, which has a residual norm of approximately 700. The Gauss-Newton algorithm likely falls into a known pitfall. The initial guess is very far from the optimal solution in parameter space, and parameter space has many local minima, local saddle points, and is shaped like a valley with trade-off between thickness/depth and the conductivity being able to both match the solution. That is, our inversion problem is *highly* non-unique. Therefore, the Gauss-Newton algorithm, while being able to push the model largely in the direction of a true solution, still did not reach a nearly weak enough conductivity with the large radius compared to the 140 km thickness assumed by the "true" model. This observation motivated our decision to pursue more advanced techniques rather than incorporating more flyby data with the same method.

The Levenberg-Marquardt Trust Region Reflective fit performs exceptionally well: the individual misfits (in units of nT) are on the order of $0.1 - 1$: that is, they are approximately the order of the noise that we have imposed. This is the best-case scenario for an inversion with noisy data. As can be seen in Figure 3, the lime green curve representing the Levenberg-Marquardt recovered dipole lines up *exactly* with the trends in the synthetic data we have generated. The sum of squared residuals, again in units of nT, is only 94. With 300 total datapoints, this yields an RMS error of ≈ 0.8 nT. Hence, the data is *slightly* overfit, but the RMS error is the same order of magnitude as the imposed noise, which is indicative of a fit with high confidence. Outputting the Jacobian indicates that the model is far more sensitive to the conductivity than it is to either radius r_0, r_1 individually, with the entries differing by 2-3 orders of magnitude. Therefore, the direction of steepest descent is a vector that is mainly pointed along the conductivity coordinate in parameter space. Finally, the optimality of the fit approached zero (final value of 0.018) as it is expected for a highly successful inversion routine: this is a measure of goodness-of-fit, equivalent to the norm of the gradient of steepest descent in parameter space. It approaching zero is a representation of finding a minimum in parameter space.

However, as discussed above, there are many non-unique solutions to a given observed signal. That is, parameter space contains many local minima. This is evident because the recovered model deviates from the input model by a substantial amount. While the conductivity differs by less than 10%, the inner and outer radii are shifted inward within the moon. The recovered ocean thickness is only slightly larger than 50 km, far thinner than the 140 km ocean prescribed. It also is at a depth of over 200 km, which is deeper than geophysical expectations place the liquid water ocean of Europa (Schubert et al., 2004). Hence, this demonstrates the high degree of non-uniqueness that we are dealing with in this application of inverse modeling. This observation motivated us to develop the statistical method, applying the Metropolis Hastings MCMC algorithm. However, we only used the synthetic data to test this algorithm, since the Levenberg-Marquardt routine was able to achieve a highly strong fit to the data. Non-synthetic data observed along a real flyby of Europa will contain a far more substantial degree of systematic error stemming from, e.g., non-stationarities in the ambient plasma, the plasma interaction (pileup and draping), slight rotation of the background magnetospheric field vector $\mathbf{B}_0$. For this reason, it is imperative to apply statistical methods in tandem with the nonlinear deterministic methods we used here because the deterministic solution will not be able to perform nearly as well with those sources of systemic error.

3.2 Results for the E14 Galileo Flyby of Europa

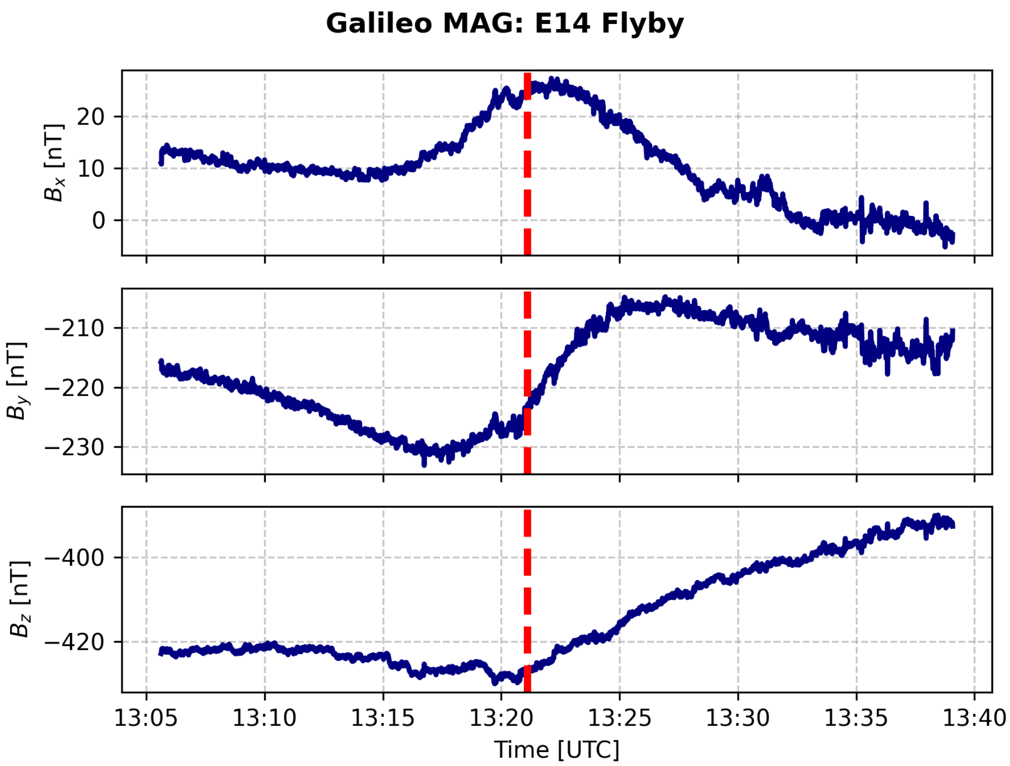

Figure 4. Data collected by the Galileo Magnetometer data along the E14 flyby of Europa (e.g., Kivelson et al., 2009). The data, shown in navy curves (separated by component), are collected at a cadence of 3 samples per second. The duration of the flyby was approximately 50 minutes, with the vast majority of the induction signal and plasma effects being observed in a ±15 min window around closest approach.

The E14 flyby of Europa, performed by the Galileo spacecraft on March 13, 1998, flew mainly along the $(+x, +y)$ direction (Kivelson et al., 2004, 2009). That is, the spacecraft approached from the Jupiter-facing, upstream side of the moon, and flew in the direction away from Jupiter and towards downstream. The z coordinate remained approximately constant, located about $0.4 - 0.5R_E$ above the $z = 0$ plane. The moon was near its maximum distance below the Jovian magnetospheric plasma sheet, where the induction signatures are expected to be strongest (due to the small phase lag, see section 2.2) and where the plasma interaction signatures are their weakest (e.g., Haynes et al., 2023, 2025). We apply a background field vector of $\mathbf{B}_0 = [8, -219.1, -406]$ nT following the approach of Harris et al. (2021). As can be seen in Figure 4, the background field was not exactly the same on inbound and outbound segments (i.e., before and after closest approach: red dashed line). For this reason, these authors linearly interpolated about closest approach. However, this will play into our inversions as a source of systemic error that we do not account for. Including a nonuniform background field vector would drastically complicate the inversion routines by requiring the storage of thousands of $\mathbf{B}_0$ values, so we defer this improved approach to the subject of a future study.

3.2.1 Nonlinear Deterministic Methods

The results obtained by our Gauss-Newton and Levenberg-Marquardt algorithms are depicted in Figure 5. As can be seen by the comparison between the recovered models (teal and cyan, respectively) and the observed data (navy), there is a substantial de-

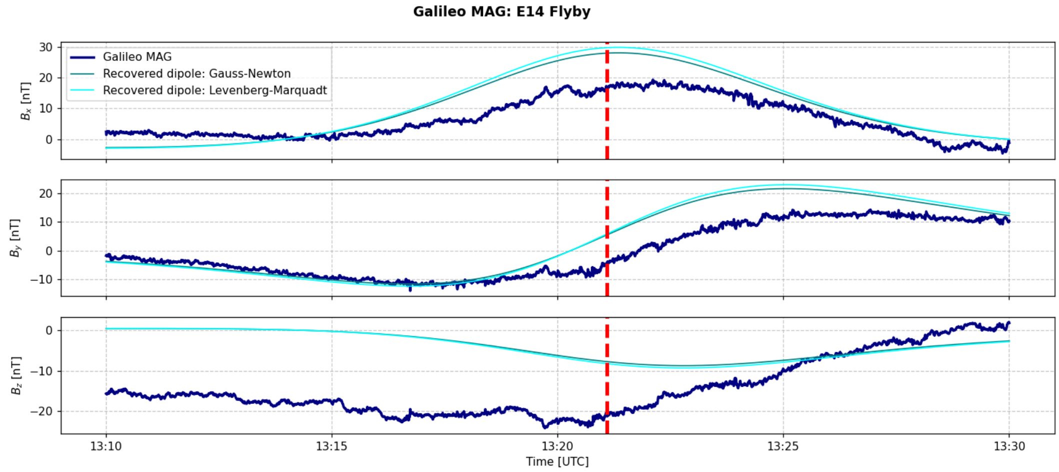

Figure 5. Comparison between (detrended, i.e., with background field $\mathbf{B}_0$ subtracted) magnetometer data measured by Galileo during the E14 flyby and recovered induced field response models for Europa's interior. The Gauss-Newton method (labeled Gauss-Newton, teal curve) recovers dipole parameters given by $\mathbf{m} = [1349.0 \text{ km}, 1550.6 \text{ km}, 2.28 \text{ S/m}]$. Levenberg-Marquardt / Trust Region Reflective (labeled Levenberg-Marquardt, cyan curve) recovers model parameters of $\mathbf{m} = [1417.4 \text{ km}, 1469.4 \text{ km}, 9.22 \text{ S/m}]$.

viation between the values of the signals just before and after closest approach. For the inversions, we truncate the data (compare timestamps in Figure 4 and Figure 5) in order to isolate the region of the flyby where the induced field signatures are most visible. The recovered model vectors for our nonlinear deterministic methods are as follows:

- $\mathbf{m}_{\text{G}-\text{N}} = [r_1 = 1349.04 \text{ km}, r_0 = 1554.64 \text{ km}, \sigma = 2.279 \text{ S/m}]$
- $\mathbf{m}_{\text{L}-\text{M}} = [r_1 = 1417.40 \text{ km}, r_0 = 1469.41 \text{ km}, \sigma = 9.22 \text{ S/m}]$

The Gauss-Newton fit, surprisingly, performs slightly better than the Levenberg-Marquardt model obtained as depicted in Figure 5. This likely stems from the fact that the Gauss-Newton method is "coarser", so it is able to approach minimums in the data residual (i.e., the objective function from equation (9)) along larger jumps and can avoid seeking out a nearer local minimum due to the fact that the Levenberg-Marquardt routine will tune the λ parameter according to the misfit. For this reason, a large, systematic misfit may impose restraints on the ability of the Levenberg-Marquardt algorithm that are not seen by the Gauss-Newton one. In addition, the models, though substantially different in parameter space, result in nearly the same dipolar signature. This again emphasizes the non-uniqueness of our problem as demonstrated in section 3.1. Furthermore, we applied an initial guess to the Levenberg-Marquardt algorithm guided by the results of our Gauss-Newton recovered model. This may have influenced the LM routine negatively. To analyze the uncertainty and quantify the "success" of our inversion strategies, we evaluate the recovered parameters, their implications, and associated confidence.

For the Gauss-Newton inversion, the recovered model suggests an ocean with a depth of ≈ 10.1 km and a thickness of ≈ 201 km. This estimate for the depth is within the range of values proposed in the literature (Vance et al., 2021b), but the thickness is slightly larger than is often assumed. The composition is not entirely constrained through conductivity, but it does pose implications for the phase and salinity of the conductive layer carrying the induced field. The conductivity of ≈ 2 S/m implies that the layer sustaining the induction signal must be a liquid or at least contain partial melt. The upper limits estimated for the conductivity of frozen layers in Europa's subsurface, even with a strong concentration of, e.g., Magnesium salts, is on the order of 10 mS/m (Zimmer et al., 2000).

Following the approach from section 3.1, the uncertainty can be assessed by again investigating the residual norm and the RMS misfit. The residual norm for the Gauss-Newton algorithm has individual misfits on the order of 0.1-10 nT, summing to about 60,000 nT. This yields a reduced Chi-squared statistic of nearly 3. This is expected given the systematic misfit between the data and the model, and it also is consistent with the model capturing approximately the behavior of the induction signal while failing to account for its magnitude around closest approach: the magnitude is once again overestimated.

For the Levenberg-Marquardt fit, the model implies a depth of 91.4 km with a thickness of only 52 km. However, the conductivity obtained by the model is roughly a factor of 4 higher than that for the Gauss-Newton inversion. This is likely due to the "trap" of there being a far higher sensitivity of the model to the conductivity parameter than either radius ($\mathcal{O}(10^{-2})$ versus $\mathcal{O}(10^{1,2})$). Hence, the steps taken by the LM algorithm were largely guided in the direction of conductivity change within parameter space, again evidenced by the gradient (i.e., the direction of steepest descent) being two orders of magnitude larger in this component. The sum of squared residuals for the Levenberg-Marquardt algorithm is roughly 87,000, yielding an RMS error of more than 4. Figure 5 indicates that there are many regions where the model does fit approximately the data, however, these are all located away from closest approach. The detrended data for the induction signal contains varying degrees of "true" induced field signatures from component to component. The B_x, B_y components are both mainly the induced field near closest approach, but modeling studies have shown that the outbound measurements of the component B_y, as well as nearly all of B_z, are influenced by signatures from Europa's plasma interaction as well (e.g., Harris et al., 2021; Rubin et al., 2015). Therefore, it is not expected for a single model of Europa's inductive response to capture the behavior in these regions and components. Also, our dataset is far larger for the E14 flyby than was generated for the synthetic flyby. This further "stresses" the deterministic algorithms and mandates a truncation / pseudo-inversion approach because the problem is extremely overdetermined (i.e., contains a more non-square G).

3.2.2 Nonlinear Stochastic Methods

We now describe the results obtained using the Markov Chain Monte Carlo method described in section 2.2.4. The distribution of results is depicted in Figure 6. As can be seen, the Markov Chain explored a decent fraction of parameter space, but remained far within the bounds imposed by the uniform prior distribution, which allowed a minimum depth of 1470 km and a maximum thickness of 200 km. The conductivity values were allowed to range over 3 orders of magnitude, however, the sensitivity to this parameter caused samples taken near either end of the prior distribution to be rejected most of the time. The recovered statistical bounds on our parameters are as follows:

- Mean (50%) solution: $\mathbf{m} = [1471.79$ km, 1549.05 km, 4.660 S/m$]$
- 10% solution: $\mathbf{m} = [1464.59$ km, 1545.18 km, 2.113 S/m$]$
- 90% solution: $\mathbf{m} = [1480.78$ km, 1552.97 km, 8.55 S/m$]$

The individual posterior distributions are included below in Figures 8, 7, 9. The uncertainty percentiles are described by the vertical dashed lines. The distributions indicate that the inner radius is more uncertain than the outer radius. This makes sense physically because of the skin depth: the electromagnetic signal penetrating to the bottom of the (sufficiently thick) conducting layer will be attenuated, so a response from deeper and deeper within the ocean is harder to "see" in the data. That is, it makes progressively weaker contributions. This is why a thin spherical conducting shell, at the driving frequencies of the Jovian field $1/T_s$, produces approximately the same response as a perfect conducting sphere. Interestingly, the inner radius takes a bimodal distribution, while the outer radius appears to be somewhat bimodal but largely normally distributed.

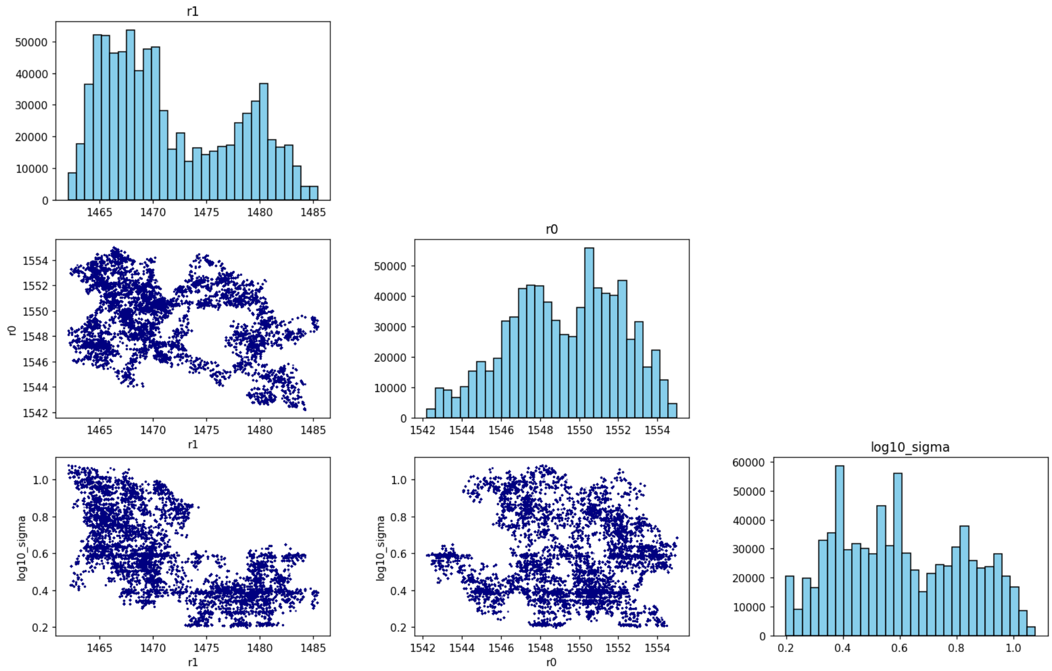

Figure 6. State distributions at the conclusion of the Metropolis-Hastings inversion algorithm. The point clouds represent different accepted states during runtime execution, and the histograms represent the obtained likelihoods. The vertical axis in the histograms is number of samples.

This could stem from the fact that at progressively increasing conductivities, a far thinner ocean is required to explain the same data. The conductivity posterior distribution is heavily skewed towards low conductivity values; this behavior is assuredly a function of the prescribed prior distribution. In order to sample as large of a space as possible, we allowed the conductivity to vary between ≈ 0.01 and 15. This high variability resulted in some samples being accepted far away from the expected conductivities on the order of 1 S/m.

To ensure the robustness of our results, we also incorporated a second run with a drastically different starting model vector $\mathbf{m}$. The recovered statistical bounds on our parameters are as follows:

- Mean (50%) solution: $\mathbf{m} = [1429.95$ km, 1539.79 km, 3.152 S/m$]$
- 10% solution: $\mathbf{m} = [1415.28$ km, 1537.35 km, 1.905 S/m$]$
- 90% solution: $\mathbf{m} = [1445.36$ km, 1543.54 km, 4.892 S/m$]$

While we do discard the burn-in data, this still had somewhat of an influence on our results. Hence, we consider both of these outputs equal representations of the likeliest states of Europa's subsurface stratification, that is, we favor the results equally. This inconsistency again stems from the highly non-unique nature of our parameter space. While the variance in radii was roughly the same, the variance in conductivity was lower. For this reason, we do favor lower conductivity values (closer to 1 S/m than 10 S/m). This is reflected in both posterior distributions and their skewness (see Figure 9). While we have several more results we could incorporate, we refrain from doing so here because of the already substantial length of this manuscript. These results emphasize the need for different datasets to be combined in order to actually constrain Europa's ocean location and composition at the level of accuracy targeted by the Europa Clipper mission.

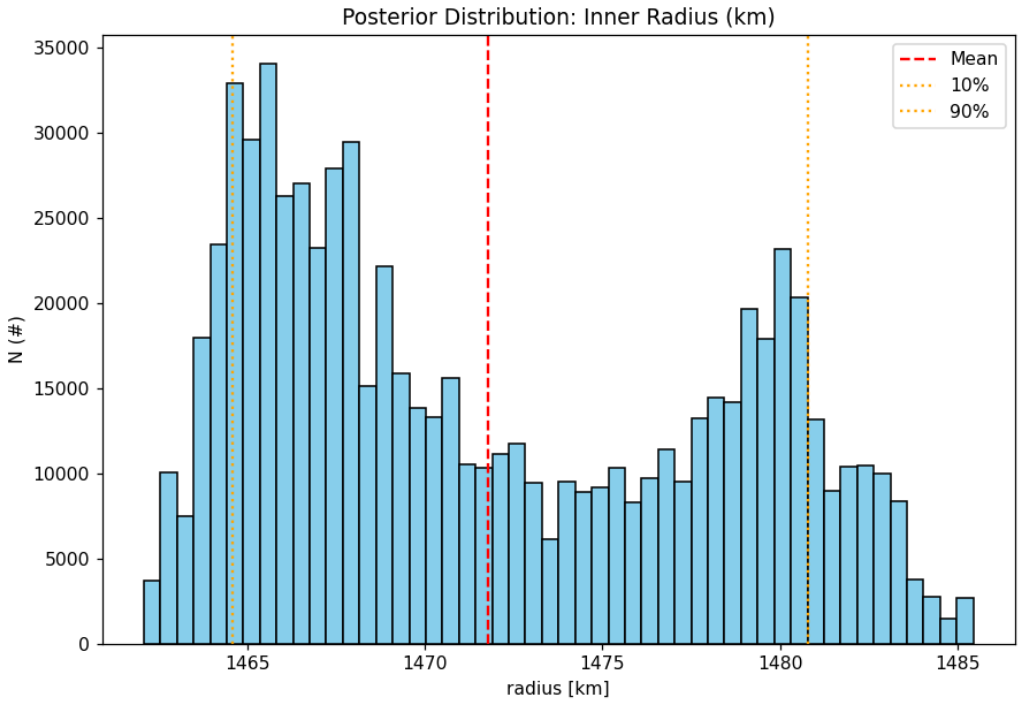

Figure 7. Posterior distribution for the inner radius r_1.

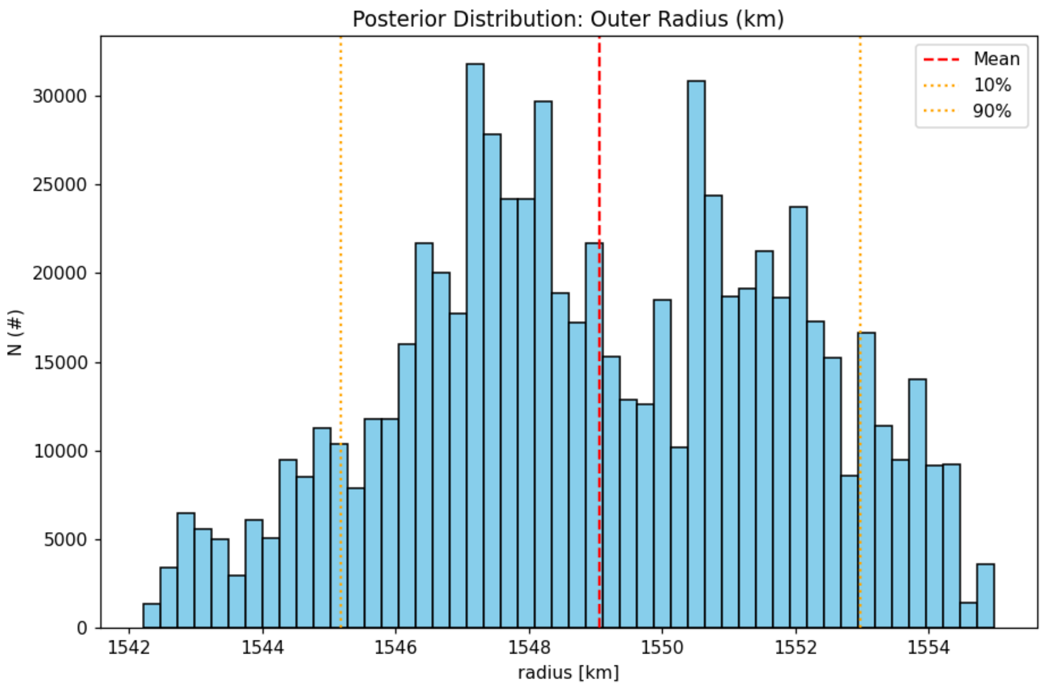

Figure 8. Posterior distribution for the outer radius r_0.

4 Conclusions

Using data from the E14 flyby of Europa, we have attempted to constrain the conductivity, depth, and thickness of Europa's subsurface ocean. Due to the highly overdetermined and non-unique nature of this inversion problem, we attempted a suite of dif-

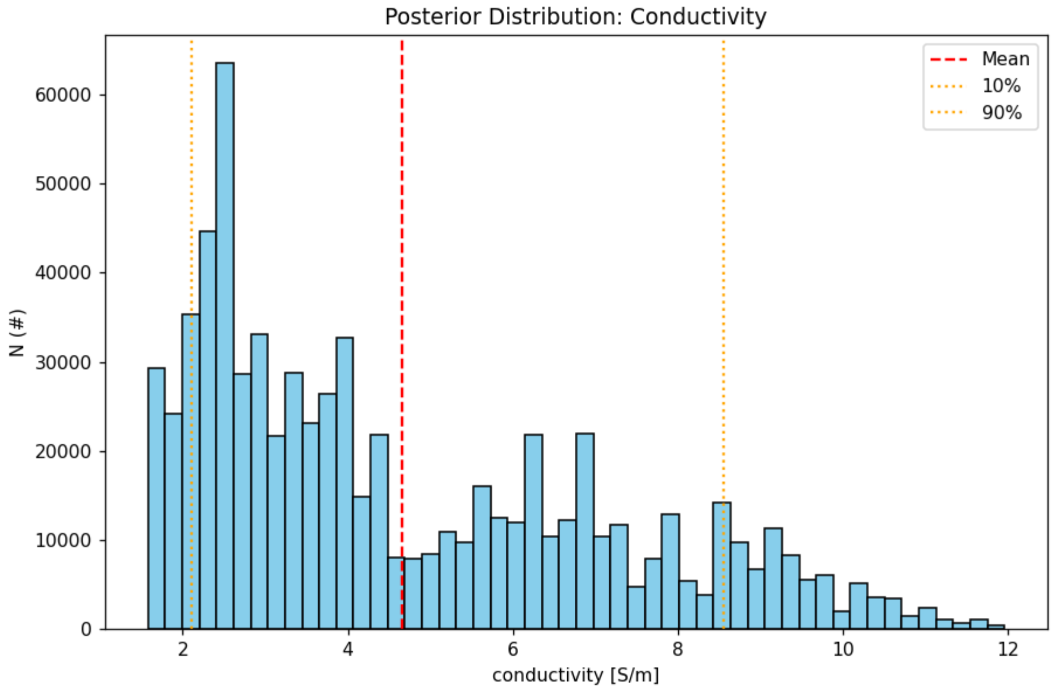

Figure 9. Posterior distribution for the conductivity σ.

ferent methods to varying degrees of success. By employing a Metropolis Hastings stochastic algorithm, we have attempted to quantify the distribution of most likely positions for the inner radius, outer radius, and conductivity of Europa's ocean. The data from the E14 flyby suggest with a high degree of confidence that the ocean layer underneath Europa's surface is of a liquid phase with a substantial concentration of magnesium salts and sulfur hydrochlorides: conductivity values of $1-5$ S/m are favored in our results, at least 2 orders of magnitude above the upper limits for a non-liquid phase conducting layer (Zimmer et al., 2000). In addition, the ocean has a favored thickness of approximately $80-120$ km and a favored depth of $8-22$ km. These parameters are consistent with geophysical modeling and induction sounding.

While our nonlinear deterministic methods are robustly implemented for "clean" data, as demonstrated by the comparison between synthetic data and recovered model in Figure 5, the deterministic approach is not suitable to constrain Europa's subsurface stratification due to the highly overdetermined, non-unique nature of our problem. Many different solutions can all match data equally well, and there are few parameters to "tune" in a simplified model that cause the full (non-truncated) inversions to fail quickly from singularities in, e.g., the Jacobian. There are approaches that can apply forward-modeling to the plasma interaction (e.g., Müller et al., 2011; Kriegel et al., 2014; Arnold et al., 2019). However, they are not tractable to incorporate into any Bayesian sampling methods, and must be reserved for an attempt at noise removal. In addition, there are non-stationarities in the upstream conditions at Europa that are impossible to constrain without multiple satellites in the moon's vicinity. For this reason, we conclude that a statistical approach may constrain the likelihood of different states for Europa's subsurface structuring, but that it is impossible to determine the precise nature of the moon's subsurface stratification without multi-point observations. For instance, a magnetometer deployed to Europa's surface could provide local driving frequencies that differ substantially from the natural ones that arise in Jupiter's environment. Additionally, the Europa Clipper spacecraft will have a microwave radiometer and be able to measure Europa's movement precisely with altimetry and gravity measurements (Cangahuala et al., 2025; Levin et al., 2026;

Petricca et al., 2026). Therefore, while the magnetic induction approach may not be suitable to diagnose Europa's ocean depth, thickness, and composition in the near future, it is possible that a combination of approaches using stochastic modeling may provide the insight needed to deem the Clipper mission a "success".

Future methods that could improve the ability of the inversion to constrain Europa's subsurface structure include trans-dimensional Markov Chain Monte Carlo approaches. These algorithms incorporate the model itself into the objective function and will randomly create and annihilate model parameters based on their weighting. This yields a posterior distribution for the necessary model parameters to describe the physical observations, and would provide a potential means to separate the ambiguity found in our magnetic sounding study between the thickness $(r_0 - r_1)$, conductivity, and phase lag (not treated here, see section 2.2). Perhaps this approach could be merged with one that includes a multi-layer structure of Europa: this would allow for the calculation of a distribution of most likely subsurface structures, including a weakly conductive mantle, multiple conducting liquid layers, as well as their depths and thicknesses.

It would also improve the approach to incorporate more carefully-selected data. By incorporating a rolling average, one could potentially remove some of the influence of short-scale fluctuations due to, e.g., plasma waves in the vicinity of Europa (Shprits et al., 2018) that cause noise in the data (see Figure 4). However, this would not mitigate the effects of the plasma interaction on providing systematic deviations between a "pristine" induction signal and the field vectors observed in the moon's vicinity. This would also address the highly overdetermined nature of the problem: including on the order of $10,000$ datapoints and only 3 model parameters results in skewed sensitivities present in the Jacobian and the ability for singular matrices to arise when performing the computations (see sections 2.2.2, 2.2.3). By downscaling this dataset to a much smaller number of datapoints that carry mostly the same information, it would potentially relax the burden on stabilizing techniques that were required in this study to obtain meaningful solutions. In addition, it would facilitate incorporating the data from multiple flybys into a single inversion framework. By selecting a few flybys or windows during flybys to include in the inversion together, the systematic effects of the plasma interaction may become more "random" (i.e., noisy) compared to systemic bias.

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